

Soham Gunturu

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EDUCATION

Georgia Institute of Technology

Expected Graduation: May 2027

GPA: 4.0 / 4.0

- *B.S. in Computer Science – Intelligence and Systems & Architecture Threads*
- Organizations: Trading at GT, AI at GT, Data Science at GT, GROWER Lab, Hexlabs, Ganesh Lab, Product @ GT
- Coursework: Systems and Networks, Intro to AI, Probability and Statistics, Discrete Math, Computer Architecture, Data Structures & Algorithms, Algorithmic Design, OOP

WORK EXPERIENCE

Microsoft

Atlanta, GA

Product Fellow

Aug 2025 - Present

- Ideating and prototyping an automated incident management system using multi-agent AI in collaboration with Product @ GT.
- Estimated to save ~\$500K in development efforts annually by streamlining ticket routing and M365 incident escalations.

Amazon

Tempe, AZ

Software Development Engineer Intern

May 2025 – Aug 2025

- Built internal tooling using Java, React, and AWS to find and resolve conflicting product offers for the AmazonCustom team, reducing error diagnosis time by 20% and improving resolution scalability.
- Developed a full-stack feature to enable sellers to download sample fulfillment zip files from buyer previews, inducing a 22% increase in seller adoption of the Buyer Preview feature on the Seller Central website.
- Enhanced deployment pipeline speed by 30% by optimizing test suites and researching root issues in previous tickets.

Vijay Ganesh Lab

Atlanta, GA

Undergraduate Researcher

Apr 2025 – Present

- Leveraging SAT solvers alongside computer-algebra systems (CAS) to push the boundaries of quantum information science, with an emphasis on characterizing and optimizing Greenberger–Horne–Zeilinger (GHZ) states and solving Krenn's Conjecture.
- Researching advanced SMT techniques for formal verification and stress-testing of large-scale AI and software systems.

GROWER Lab

Atlanta, GA

Team Lead & Data Engineer

Aug 2024 - Present

- Evaluating different machine learning and threshold-based methods to aggregate power outages in existing data by contrasting with ground truth, partnering with national labs like Sandia and Oak Ridge to better understand how to approach outage analysis and policy change.
- Leading research to understand the impact of physical and cyber-attacks on US energy infrastructure through data analysis and visualization.
- Developing different machine learning algorithms to predict future attacks and/or outages in the US power grids, considering 15+ variables.

Nationwide Children's Hospital

Columbus, OH

Data Science Intern

June 2023 – Dec 2023

- Analyzed spatial datasets to identify differences between primary and metastatic tumors, resulting in a 20% faster differentiation which contributed to more targeted cancer treatments.
- Performed various bivariate correlation studies and created visualizations to understand the architecture of brain tumors
- Refined an Agent-Based Model's accuracy by 30% by fine-tuning PyTorch parameters, allowing researchers to better simulate cell interactions.

PROJECT EXPERIENCE

GT Marketplace | Typescript, NextJS, ReactJS, Firebase, Python, ExpressJS, NodeJS, HTML/CSS, OAuth2

Apr 2025 – Present

- Designed and implemented a full-stack campus marketplace app enabling students to buy, sell, and trade items with other GT students.
- Includes real-time chats, firebase authentication, and a smart recommendation system for students to dynamically explore the marketplace.

Georgia Tech RAG Chatbot | Python, NIM, Docker, ReactJS, PostgreSQL, Scrapy

Aug 2024 – Dec 2024

- Developed a RAG-based LLM in collaboration with AI@GT and NVIDIA, allowing prospective Georgia Tech students/families to easily get accurate information about the institute.
- Built a robust vector embedding pipeline using Scrapy to scrape relevant data and PostgreSQL to store and manage over a million vectors.
- Deployed a React-based application with interactive UI elements, enhancing user experience when using the chatbot.

DeepQuote | C++, Python, PyBind, PyTorch, Gymnasium, Numpy, Stable-Baseline3

Jun 2025 – Present

- Building a high-frequency trading simulator processing 10,000+ orders/sec with order flow analysis, order book mechanics, and other features.
- Using reinforcement-learning-based AI agents to compete in this simulated market, achieving 18-dimensional state spaces and supporting cutting-edge trading strategies with real-time PnL tracking.

SKILLS/INTERESTS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL, HTML, CSS, Assembly

Frameworks: Node.js, React.js, Next.js, Express.js, FastAPI, Django, Flask, React Native, Bootstrap, Dagger, Spring

Software: PyTorch, Transformers, Pandas, Scikit-Learn, PostgreSQL, Firebase, AWS, MongoDB, Docker, Scrapy, Gymnasium, Redux

Awards: Ohio Governor's Scholar, Hack OH/O 1st place, Huntington Scholar, Presidential Gold Award, Presidential Undergraduate Research Award

Leadership: Events Executive of Trading at GT, President of Taal Tadka A capella, Team Lead at GROWER Lab,

Interests: Technology, Basketball, Weightlifting, Cooking, Travel, Poker, Music